Expanding the ASEAN Digital Integration Index

Agnieszka Choczyńska

June 2026

1 Introduction

Digitalisation has become a central part of economic development (Myovella et al., 2020; Mubarak et al., 2020), crucial not only for the ICT industry, but almost every branch of the modern economies, intertwining with social and environmental aspects (Jovanović et al., 2018). It is, however, difficult to measure digital progress for several reason. First, it is multi-faceted, ranging from hard infrastructure to legal environment or human skills. Bruno et al. (2023) noted that most concepts of digital progress indicators include components of ICT infrastructure, skills and usage, but there is a huge variability in which particular metrics are used to describe them (Kotarba, 2017). Second, fast page of technological progress can make some indicators obsolete. For example, Myovella et al. (2020) showed that fixed broadband internet is less important to the developing Sub-Saharan countries, while the mobile internet has a considerable impact on economic growth. Third, there is a problem of digital divide, meaning that particular groups of people or countries may not participate in the digital progress at the same level (Billon et al., 2010; Lythreatis et al., 2022; Liao et al., 2022). This inequality can deepen the disparities in other aspects of life (Li, 2022).

On top of that, there is no widely acclaimed method to define or measure digital development. Countries and international organisations collect hundreds of different indicators, and many construct composite digitalisation indexes. There are, among others, the Digital Economy and Society Index in European Union (European Commission, 2022a), ASEAN Digital Integration Index for South-East Asian countries (ASEAN, 2023), the Digitalisation Index (DiGiX) created by BBVA Research (Cámara and Tuesta, 2017), ICT Development Index made by the International Telecommunication Union (International Telecommunication Union, 2023), Digital Intelligence Index (DII) created by the Fletcher School at Tufts University (Chakravorti et al., 2020), and many more. Each of them has a unique approach to digitalisation and uses different measures, making the cross-sectional studies almost impossible (Choczyńska et al., 2024). Researchers interested in wider comparable analyses resort to using a selection of simple indicators (Sey, 2021) or construct their own indicators (Siuta-Tokarska et al.; Fura et al., 2025).

I propose an alternative approach by reproducing the ASEAN Digital Integration Index (ADII) outside of the ASEAN region. ADII index covers a broad definition of digital development, including institutional and infrastructural readiness, environment of innovation, human skills, data protection and cybersecurity, and digital trade, payments and logistics. Its benefit lies in the use of open and international data sources (such as World Bank, UNESCO, ITU Data Hub, among others), which cover many countries outside of the region, offering high quality and reliable data. I managed to construct the index for 108 countries, out of which 14 were included in the original ADII report. For these 14 countries, the main ADII score provided 0.98 correlation with the original values, with 7.56% average error.

However, there were some limitations. Despite the promising quality of the final score, the approximation of particular pillars was of lower quality, and in most cases underestimated the scores provided by the original report. Moreover, I noticed that the errors were particularly large for smaller economies, such as Laos and Cambodia. The problems likely stem from the lower data availability of some indicators (which had to be patched with other sources) and the lack of clarity about indicators composition in the original report.

Nevertheless, compared with other commonly used composite indices, the reconstructed ADII demonstrated quite a high similarity. It is highly similar to the Digital Intelligence Index, with 0.97 Spearman's correlation. The correlations with DESI were lower, and the ITU Digitalisation Index differed the most. Despite different indicators used to construct these indices, the choice of a particular one should not much alter the results, especially in studies based on ranking countries.

Additionally, I followed Bruno et al. (2023) approach and constructed a version of ADII with reduced dimensionality. The new version is based on 24 (instead of 30) indicators, which allows to expand the dataset with 16 additional countries, while maintaining 0.99 correlation with the reconstructed ADII scores and 0.95 correlation with the original values from the report. The whole dataset is provided in the repository.

The contribution of this study is three-fold. First, it reconstructs the ADII scores for 108 countries (124 in the version with reduced dimensionality), offering other researchers a comprehensive dataset with wide geographical coverage. Second, it analyses the comparability of several composite digitalisation indices and shows that despite differences in composition, they provide very similar rankings. Finally, it demonstrates that dimensionality of digitalisation indices can be reduced with a little loss of accuracy, allowing for omission of variables with too many missing data points. These contributions may be useful for researchers selecting a digitalisation index for larger spatial studies.

2 Literature Review

2.1 Measuring digitalisation

Vial (2019) identified as many as 23 definitions of digital transformation in the literature. On top of that, there are over 100 metrics related to particular facets of this transformation, and the level of standardisation is moderate (Kotarba, 2017).

Fura et al. (2025) emphasised that no single measure can accurately represent the digital transformation. As access to the internet becomes more and more common, it stops distinguishing advanced digital economies from the ones that lag behind (Liao et al., 2022). There is a need to consider other aspects, such as digital skills and education, technology adoption by businesses and public sector, or cybersecurity and legal measures.

This is why governments, researchers and organisations resort to constructing composite digitalisation indexes, aiming at capturing the complexity of the subject through collection of tens of simpler metrics. Perumal et al. (2024) performed a review of these indicators used in academic studies, finding that the ICT Development Index was the most common, despite being quite narrowly focused on the infrastructure and internet access. Bruno et al. (2023) noted that composite indicators differ in details, but typically cover three main areas: ICT infrastructure, ICT skills, and ICT usage and adoption.

Nevertheless, they are not interchangeable. Choczyńska et al. (2024) compared three composite indexes: DESI, ADII and DII (mention in the introduction!), finding that almost none of metrics constituting them repeated in other indexes. They compared ranking of countries based on DESI and DII (in Europe) and ADII and DII (in ASEAN region), showing that the correlation is quite high, but the choice of index would still affect which country would be deemed as more digitally advanced.

Another problem can be the number of indicators used in the composite indices, which can be highly correlated with each other (Bruno et al., 2023). Apart from multicollinearity, this may unnecessarily limit the geographical scope, increasing the likelihood of a country having at least one missing data.

2.2 The ASEAN Digital Integration Index

The ADII was created by the ASEAN Economic Community in 2020-2021 as part of the efforts to strengthen and integrate the digital development of the region in the face of the COVID-19 pandemic. In 2022-2023, it was updated to version 2.0 to provide a comprehensive update on the digital transformation of the ASEAN Member States. The index consists of six pillars, representing different domains of digital development, but focusing strongly on the role of technology in the economy.

The indicators making up the index were selected based on six key criteria:

- Relevance - Data must be clear, accurate, and directly aligned with Digital Integration Framework priority areas.
- Accessibility - Data must be easily obtainable and made available without cost.
- Coverage - Indicators must encompass at least eight of the ten ASEAN Member States.
- Timeliness - Datasets must be recent, typically published from 2017 onwards.
- Consistency - Data sources must be published on a regular and predictable basis.
- Data must originate from reputable sources with open methodologies and disclosed limitations.

Although the ADII was created with ASEAN states in mind, these selection criteria make it suitable for adoption in other parts of the world. Most of the data is sourced from global, publicly available sources, making it easier to extend outside of the region. This is in contrast to the DESI index, in which some indicators are collected solely for this purpose by the EU Member States institutions. On the other hand, although selection criteria included Timeliness and Consistency, institutions providing the data may discontinue their selection. For example, several indicators come from the World Economic Forum's Global Competitiveness Report, which was not updated since 2019.

The six pillars of the ADII map precisely to the priority areas of the ASEAN Digital Integration Framework (DIF), which regional leaders have identified as essential for a unified digital economy. Each pillar consists of 5 indicators, although some of them may be aggregations of several simpler measures. The indicators are transformed to a scale between 1 to 20, and each pillar is the sum of the maximum value of 100. The final index is the average of all its pillars.

Pillar 1: Digitally Enabled Trade & Logistics - Assess the adoption of digital technologies and infrastructure used to facilitate seamless cross-border trade. The digitalisation of this sector may be less important for economies less reliant on international trade. However, part of the indicators can relate to transport and trade of goods within the country.

Pillar 2: Data Protection & Cybersecurity - Measure the degree to which frameworks protect data privacy while encouraging secure digital innovation. This is an important safeguard ensuring the fair adoption of digital technologies and a big benefit over other popular indicators (such as European DESI) that do not include cybersecurity indicators.

Pillar 3: Digital Payments & Identities - Assess the adoption and interoperability of digital financial services and national identity systems. Digital payments have been quickly growing in popularity in the region, as e-commerce is the main area of ASEAN's digital economy (Rumata and Sastrosubroto, 2020).

Pillar 4: Digital Skills & Talent - Quantify the availability of a skilled workforce capable of driving the regional digital economy. The possibility of actually making use of digital advancements and quickly switching to new technologies in the case of a crisis is dependent on human skills and attitudes. Moreover, education and knowledge is a crucial area due to its contribution into the digital divide (Lythreatis et al., 2022).

Pillar 5: Innovation & Entrepreneurship - Examine the business environment's ability to foster and scale digital start-ups and IP creation, the availability of financing and ease of starting a business.

Pillar 6: Institutional & Infrastructural Readiness - Determine the public sector's responsiveness to digital change and foundational infrastructure availability. This pillar includes both hardware infrastructure and laws and institutions.

The indicators included in the ADII come from seven sources, listed in Table 1.

The **OECD Trade Facilitation Indicators** is a database maintained by the Organisation of Economic Cooperation and Development. It contains information on movement, release, and clearance of goods at the border in 163 countries. The **World Bank Open Data** is the source maintained by the World Bank Group, providing a wide selection of global development data. From this source, ADII uses the Logistic Performance Index, Global Financial Development database, and Identification for Development database. The World Bank also provides access to Global Competitiveness indicators by the World Economic Forum. Another global source is the ITU Global Cybersecurity Index, created by the **International Telecommunication Union**. These data are supported by the Data Protection

Index from the 2020 report prepared by the **TRPC**. It was, however, only collected for 30 countries, mainly from the ASEAN. ADII also sources indicators from the **UNESCO Institute for Statistics** database, the Global Innovation Index report by **WIPO**, and the **United Nation's** Global Survey on Digital and Sustainable Trade Facilitation.

The ADII was released in two editions, the first in 2021, and the second in 2023. In the first edition, pillars 2 (cybersecurity) and 6 (institutional and infrastructural readiness) received the highest scores. But in the second edition, ASEAN countries rapidly improved scores in the first pillar (related to digital payments and identities), probably due to e-commerce's increasing popularity after the pandemic. Pillar 4 (digital skills) has been and remains the least developed area, while Pillar 5 (innovation and entrepreneurship) is only sloghtly better.

3 ADII reconstruction

3.1 Data collection

First, I identified the indicators within the sources and collected the data. The sources were either in a form of database, where downloading the data was straightforward, or a report, where I scraped the data using NotebookLM tools. Table 1 details sources and issues encountered during the collection exercise. In a few cases, the source contained several indicators that could match the description in ADII report, and it was not clear which should be included.

Indicator 1.1 is a compound index describing how trade and customs processes are supported by digital technologies. Among the OECD Trade Facilitation Indicators, I identified several indicators that matched this description and selected five with enough data coverage: G4 (electronic pre-arrival processing), G6 (Processing system electronic payment), G9 (electronic data interchange), G10 (automated processing - goods release) and G12 (full-time automated processing). For the Indicator 1.3 I chose variable A7 from the OECD Trade Facilitation Indicators database, which describes Import/Export procedures.

According to the ADII report, indicator 3.4 should come from the World Bank's Global Financial Development (Findex) database, but I found this variable in the World Bank's Identification for Development database. Given that this is the same institution, it could have been a simple mistake in the report. Another indicator from this database (3.5) is supposed to describe the degree to which a digitized ID system is in place. I found two variables regarding digitised ID, one for the percentage of the population using it, and the second being a binary variable with 1 if the country has a digitised ID and 0 otherwise. I chose the latter because of much better data availability, although none of them seems to directly match the description.

In three cases, the original dataset was very limited, and I used other sources to expand it. The indicator 5.5 (protection of intellectual property) was available for only 53 countries. When missing, I supplemented it with the same variable from the World Bank database [detailed name]. When indicator 3.2 (using digital technologies for financial transactions) was not available, I copied the value for the country from the indicator 3.1 (using digital technologies for banking), because digital banking could be categorised as part of the wider financial uses. However, it can mean that indicator 3.2 is underestimated in the dataset.

The indicator 6.1 (active mobile-broadband subscription) was taken from the same source, but from the ITU Data Hub database instead of the report. For indicator 6.2 (proportion of internet users), I found the same variable from the World Bank database.

The most difficult case was with the indicator 2.1 (data protection), originally sourced from the TRPC report which only included 30 countries. I reconstructed this variable by using the DLA PIPER database of legal data protection measures and the questionaire included in the original report. This could introduce some bias, as the the protection measures may have changed through the years. [check how similar it is for the countries I do have]

Tabela 1: Sources of data and issues encountered during the collection process.

Indicator	**Description**	**Source**	**Issues**
1.1	Degree to which trade/customs processes are supported by digital technologies	OECD Trade Facilitation Indicators	Not sure which measures are aggregated into this indicator
1.2	Degree to which digital certificates and signatures are in place	OECD Trade Facilitation Indicators	
1.3	Degree to which international standards for trade documents and procedures are followed	OECD Trade Facilitation Indicators	

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Indicator	Description	Source	Issues
1.4	Level of quality of trade- and transport-related infrastructure	World Bank Logistic Performance Index	
1.5	Level of competence and quality of logistics services	World Bank Logistic Performance Index	
2.1	Degree to which data protection measures are in place	TRPC Data Protection Index	TRPC only provided data for 30 countries, reproduced the rest
2.2	Level of legislative and regulatory cybersecurity capabilities	ITU Global Cybersecurity Index	
2.3	Level of institutional cybersecurity capabilities	ITU Global Cybersecurity Index	
2.4	Level of technical cybersecurity capabilities	ITU Global Cybersecurity Index	
2.5	Level of international cooperation on cybersecurity	ITU Global Cybersecurity Index	
3.1	Proportion of people who use digital platforms or devices for banking purposes only	World Bank, Global Financial Development (Findex) database	

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Indicator	Description	Source	Issues
3.2	Proportion of people who use digital platforms or devices for any type of financial transaction	World Bank, Global Financial Development (Findex) database	If not available, used 3.1
3.3	Degree to which legal frameworks enable electronic transactions	UN Global Survey on Digital and Sustainable Trade Facilitation	Got the same indicator from OECD Trade Facilitation Indicators
3.4	Proportion of people who have a national identity card	World Bank, Global Financial Development (Findex) database	Found this in: World Bank, Identification for Development
3.5	Degree to which a digitized ID system is in place	World Bank, Identification for Development	Only found a binary variable
4.1	Proportion of graduates in science, technology, engineering, and mathematics	UNESCO UIS database	
4.2	Proportion of employment in knowledge-intensive services	WIPO, Global Innovation Index (GII) database 2023	
4.3	Level of multi-stakeholder collaboration in R&D	World Economic Forum WEF), Global Competitiveness Report	

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Indicator	Description	Source	Issues
4.4	Degree to which the active population has digital skills	World Economic Forum WEF), Global Competitiveness Report	
4.5	Degree to which graduates have business-relevant skillsets	World Economic Forum WEF), Global Competitiveness Report	
5.1	Degree to which venture capital is available	World Economic Forum WEF), Global Competitiveness Report	
5.2	Proportion of GDP expenditure on R&D	WIPO, Global Innovation Index (GII) database 2023	
5.3	Degree to which innovative companies can grow	World Economic Forum WEF), Global Competitiveness Report	
5.4	Degree to which it is easy to start a business	World Bank, Ease of Doing Business Index	
5.5	Degree to which intellectual property protection frameworks are in place and are enforced	Global Innovation Policy Center (GIPC), International IP Index (IIPI)	53 countries only, but substituted with the same variable from the World Bank

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Indicator	Description	Source	Issues
6.1	Proportion of Active mobile-broadband subscriptions	ITU, Measuring the Information Society Report 2018	Got data from ITU Data Hub from 2018 or before
6.2	Proportion of Internet users	ITU, Country ICT Database	Did not find the data, used the same variable from the World Bank
6.3	Degree to which government services are available and accessible digitally	UN e-Government Knowledge base	
6.4	Degree to which a government is considered responsive to disruption and change	World Economic Forum WEF), Global Competitiveness Report	
6.5	Degree to which a legal framework is considered conducive for digital innovation	World Economic Forum WEF), Global Competitiveness Report	

3.2 Constructing the scores

Following the procedure described in the ADII report, each variable was rescaled to take values between 0 and 20 by dividing it by the 'top value' and multiplying by 20. In most cases, the indicators were initially created in a pre-defined scale, and the 'top value' was the maximum. If the variable was in percentages, the top value was set to 100. Indicator 6.1 (mobile-broadband subscription per 100 people) did not have a maximum value because one citizen can have more than one subscription, and the value exceeded 100 in many developed countries. I could set the 'top value' at the maximum available in the dataset (as it was done in the ADII report), but this solution had some drawbacks. First, the distribution was highly skewed and the resulting indicator would have a majority of the countries with very low scores, even if almost all citizens had subscriptions. Second, it would not reproduce the scores in the ADII report, because the maximum value among just the ASEAN sample would be different than in the whole dataset. Finally, I capped the values at 100 and set that as the top value for rescaling.

Each pillar is constructed as a sum of five underlying indicators, so the pillar scores fall naturally on a scale of 0-100. The final index is computed as an average from six pillars, preserving that scale. In the ADII report, if an indicator was missing, the pillar was computed by summing other indicators and rescaling. Since I had many countries with too many missing variables, I chose only those with at most three indicators missing. This way more countries could be included with the integrity of scores preserved as much as possible.

3.3 Comparison with the original scores

Finally, I was able to construct the ADII scores for 108 countries. Out of 16 countries from the original reports, 14 are included in my dataset, and only the data for Brunei Derussalam and Myanmar was missing. I compared the pillar scores and found an average of 12.3% error, with the biggest disparities for smaller economies (e.g. 33.6% for Laos and 22.3% for Cambodia). More details of the comparison can be found in Table 2. The correlations between original and reconstructed pillar scores are high, with the lowest of 0.78 for pillar 4 (digital skills), and the correlation for the aggregated index reaching 0.98. The average errors (where errors are not treated as absolute distance between original and reconstructed value, but as simple difference) show that the reconstructed scores are usually smaller than the ones obtained by ASEAN. The only exception was the first pillar, with a slightly negative average error.

The analysis of mean percentage error shows that the biggest discrepancy is in the Pillar 3 (digital payments and identities), where the reconstructed scores underestimated values reported in the ADII report. This pillar includes several indicators that I had trouble to match (3.3 and 3.5), and indicator 3.2 with a very limited data coverage. This is likely what is causing the issues.

4 Results

4.1 Digitalisation across the world

The full table of country scores can be found in the Appendix. The highest scores belonged to Singapore (86.5), Finland (85.9) and Denmark (83.6), followed closely by the United States (83.5), showing the the digital frontiers could developed on many continents. However, Figure 1 and Table 3 show that in regional analysis, Europe had the highest average and the most

Pillar	Correlation	Average pct. error	Average error
P1	0.96	10.14	-0.63
P2	0.97	7.01	1.14
P3	0.86	22.62	9.52
P4	0.78	12.35	4.38
P5	0.86	13.04	5.36
P6	0.93	8.86	5.55
Index	0.98	7.56	4.22

Tabela 2: Comparison of recreated pillar scores with original values from the ADII report (ASEAN, 2023) N = 14.

equal scores, while the Americas and Asia are very diverse in the level of development. African countries scored the lowest across all pillars, but also with mostly small standard deviations, with the exception of Pillar 2. Note that due to the fact that the dataset contains only two countries from Oceania (Australia and New Zealand), they were included in Asia, which could have slightly overestimate the regional mean.

Figure 1 indicates as well that the countries with missing data concentrate in Africa, Middle East, Central Asia and Eastern Europe. Therefore, the regional comparisons should be interpreted with caution. Lack of data from developing nations is a big problem in global digitalisation studies, as this is both where the transition could make a big change to the economy, and where the digital divide is likely to be deeper (Mubarak et al., 2020). However, these countries often do not yet have the institutions or culture of collecting and openly sharing the data (Oloyede et al., 2023).

4.2 Comparison with other composite indicators

Next, I want to compare the scores of the reconstructed ADII indicator to other composite indexes. High similarity is a beneficial characteristic, because it means that the underlying concept of digitalisation is robust and the choice of specific indicator does not substantially alter study results. To assess the similarity I will use Pearson's and Spearman's correlations.

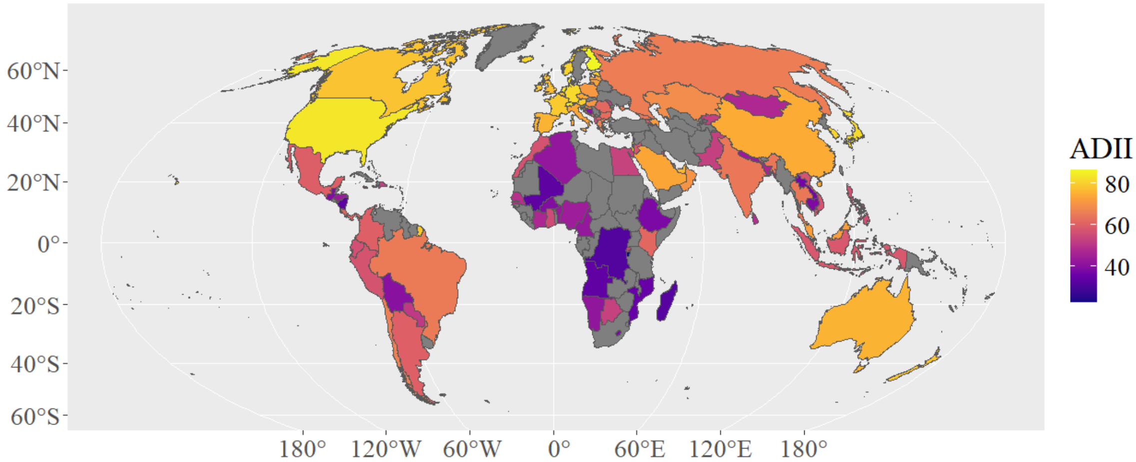
Pearson's correlation if given by the equation:

$$R_p = \frac{\sum_i^N [(x_i - \bar{x})(y_i - \bar{y}))}{S_x S_y} \tag{1}$$

where x_i is a value of ADII for the country i, y_i is the value of the other indicator for that country, $\bar{x}$ is a arithmetic mean of the scores, and S_x is their standard deviation. I use the Pearson's correlation because of its relation to the linear dependency between the variables

Region	All	Africa	Americas	Asia and Oceania	Europe
N	108	24	18	33	33
Pillar 1	66.0 (20.7)	41.70 (11.10)	67.70 (16.60)	68.1 (18.8)	80.7 (13.4)
Pillar 2	82.9 (19.5)	74.00 (23.10)	72.80 (20.00)	85.2 (18.9)	92.7 (9.91)
Pillar 3	66.0 (27.6)	36.40 (19.50)	65.10 (22.30)	66.5 (24.0)	87.5 (17.4)
Pillar 4	41.7 (9.73)	33.70 (6.23)	37.1 (8.19)	44.1 (8.94)	47.6 (8.45)
Pillar 5	46.7 (14.6)	37.50 (8.40)	38.4 (13.3)	49.6 (14.1)	55.0 (13.7)
Pillar 6	59.0 (16.9)	39.00 (10.80)	56.2 (13.0)	63.7 (14.8)	70.5 (9.97)
Index	60.4 (15.8)	43.70 (10.50)	56.2 (13.1)	62.9 (13.7)	72.3 (10.4)

Tabela 3: Average scores per region, standard deviation in the brackets.



Rysunek 1: Reconstructed ADII scores. Countries with unavailable data are in grey,

in classical linear regression models, which are commonly used by researchers. A score close to 1 would mean that two composite indices convey almost the same information and could be used in the model interchangeably. Otherwise, the dependencies found by the researcher may be specific to a particular index, and not the phenomenon of globalisation itself.

Spearman's correlation can be computed with the same Equation 1, but using ranked indicators instead of original scores. A value close to 1 would indicate that two composite indexes point at the same order of countries from most to least digitally advanced. If the Spearman's correlation is too low, studies based on comparison to global or regional frontiers may be affected by the choice of specific index.

I test three composite indices: DESI, IDI and DII. DESI (Digital Economy and Society Index) is a digitalisation measurement standard for the European Union, collected since 2014 (European Commission, 2022b). It covers many areas of digital transformation, with a strong focus on public services, human skills and the technological adoption in business. Apart from monitoring the digital progress of the EU (Żmija et al.), it was used to measure the Member States digital convergence (see e.g. Borowiecki et al. 2021; Kovács et al. 2023), suggesting optimal policies for development (Bruno et al., 2023).

IDI (ITU Digitalisation Index) was found by Perumal et al. (2024) to be the most commonly used in research papers. Its relatively simple structure consists of ... indicators measuring the availability and use of ICT technologies (International Telecommunication Union, 2023).

DII (Digital Intelligence Index) was created by the Fletcher School at Tufts University (Chakravorti et al., 2020). The idea behind the pillars is quite different than in the case of DESI and ADII, as they represent the Supply and Demand sides of digital economy, as well as pillars for Institutions and Innovation components. Unfortunately, the latest available scores are from the 2020, but I include this index to replicate the results from Choczyńska et al. (2024), where the Spearman correlation between original scores of ADII and DII for 13 Asian countries was 0.93.

The results indicate quite good similarity between composite indices. There were 24 countries with available data for both indices, the Pearson correlation was 0.87 and the Spearman correlation was 0.89. IDI had the widest coverage, with scores available for almost all countries in my ADII dataset (105 out of 108 countries). However, the correlations are lower: Pearson's was 0.83 and Spearman's was 0.87. DII was available for 75 countries from the ADII dataset, and the correlations were, respectively: 0.94 and 0.97, showing the highest compatibility with ADII, even despite the time difference. The score being almost the same as in Choczyńska et al. (2024) proves that high similarity between these two indices was not just due to a small sample or a regional specifics.

In light of these results, despite different construction and indicators used, the selected composite indices are quite similar, especially if used to rank countries. The lowest correlations were obtained for the IDI, probably because it focuses on technical aspects of digitalisation, omitting social and economic context represented in other indices.

4.3 Dimensionality reduction

Bruno et al. (2023) pointed out that composite indexes may suffer from multicollinearity and unnecessary complexity by incorporating too many similar indicators. They proposed a procedure to reduced the number of indicators and applied it to the DESI index. Based on his approach I constructed a reduced version of ADII that allows to omit some indicators with missing values and expand the dataset to 16 additional countries.

The method is based on Principal Component Analysis (PCA). PCA captures the common components explained by correlated indicators, which is used to construct new, orthogonal variables that capture as much variance in the original indicators as possible.

Let $\mathbf{X} \in \mathbb{R}^{N \times P}$ denote the data matrix of P indicators observed for N countries. PCA is applied to centred and standardised data:

$$\tilde{\mathbf{X}}_{ij} = \frac{\mathbf{X}_{ij} - \bar{\mathbf{X}}_{\cdot j}}{s_j}, \tag{2}$$

where $\bar{\mathbf{X}}_{\cdot j}$ and s_j are the sample mean and standard deviation of indicator j.

PCA finds a direction $\mathbf{w}_1 \in \mathbb{R}^P$ for the first component by solving

$$\mathbf{w}_1 = \arg \max_{\|\mathbf{w}\|_2 = 1} \mathrm{Var}(\tilde{\mathbf{X}}\mathbf{w}). \tag{3}$$

With the sample covariance matrix

$$\mathbf{S} = \frac{1}{N-1} \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}, \tag{4}$$

the solution is given by the eigenvector corresponding to the largest eigenvalue of $\mathbf{S}$, i.e.

$$\mathbf{S}\mathbf{w}_k = \lambda_k \mathbf{w}_k, \qquad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_P \geq 0. \tag{5}$$

The proportion of variance explained by component k is

$$\mathrm{PVE}_k = \frac{\lambda_k}{\sum_{j=1}^{P} \lambda_j}. \tag{6}$$

I apply the PCA for each pillar and select three first components, which explains at least 90% of variance of the original pillar. This number varies slightly depending on the case, with the highest proportion of variance for pillar 4 and the worst results for pillar 3. Next, I analyse the correlations between original indicators and components from PCA. For the each component, I select the indicator with the strongest correlation, or the second best, if it was already chosen for the previous component. Because my goal is to recreate the ADII for as many countries as possible, I slightly modify the Bruno et al. (2023) approach by taking into account data availability. I choose a second-best indicator, if it has substantially better coverage. For example, in the pillar 5, the indicator 5.4 (ease of starting a business) had a -0.76 correlation with the second component, but was unavailable in 115 countries. Instead, I chose the indicator 5.3 (degree to which innovative companies can grow), which had correlation of 0.41, but was unavailable in only 75 countries.

The reduced version of ADII contains only 3 indicators per pillar, 24 in total, compared to 30 in the original version. Pillar scores are computed in the same manner as before and rescaled, so that the pillar remains on the scale 0-100 and the remaining indicators take up the weights from the removed ones. The final score has a 0.99 correlation with the original one, although the correlations for particular pillars remain lower: 0.98, 0.94, 0.95, 0.95, 0.96, and 0.98.

When compared to the scores from ADII report, it has 7.06 percentage error, which is 0.5 percentage point lower than than before the reduction. However, the results for particular pillars are much less precise. Table 4 with a more detailed comparison can be found in the Appendix, along with a map showing reduced ADII scores (Figure 2).

5 Discussion

This work expands on the ASEAN (2023) report to create an open composite digitalisation indicator with a world-wide range. Although the main index scores could be recreated with a satisfactory precision, the quality at the pillar level is noticeably worse. Moreover, despite timeliness and consistency criteria for selection of the original indicators, not every source continues to provides the data, which would limit the ADII in the upcoming years.

The article adds to the literature on the similarity of composite digitalisation indexes (Korzhyk et al., 2023; Choczyńska et al., 2024). Despite a wide range of metrics (Kotarba, 2017) and working definitions (Vial, 2019), Bruno et al. (2023) notes their functional similarity. Choczyńska et al. (2024) verified this in practice, comparing the rankings and correlation between DII-DESI and DII-ADII within their respective regions. Recreating the

ADII for the rest of the world allowed to repeat that analysis on a much larger dataset. The study confirmed that DII and ADII are closely related, while the similarity with DESI and IDI is lower.

This study also adds to the strand of research of dimensionality of composite indices. It follows a dimensionality reduction methodology of Bruno et al. (2023), with a slight change to prioritise indicators with wider coverage. The results show that reducing the number of indicators can help expand the dataset with several additional countries without a substantial loss of quality. This is especially important since data availability problems are more common among developing countries (Oloyede et al., 2023), often excluding them from the analysis.

6 Conclusions

Digital progress on a country level is typically measured with composite indices, that can incorporate many facets of digitalisation, from hard infrastructure to human skills or legal environment. The problem is, there are many propositions of digitalisation indexes, differing in the areas covered, selection of indicators and geographical scopes. Some of them are design for regional needs (such as EU's DESI or ASEAN's ADII), which limits wider cross-country comparisons. Moreover, the question of whether the choice of a particular index to represent digitalisation can impact the outcomes of a study is not yet well researched.

The idea behind this paper is to reconstruct the ASEAN's ADII index for the rest of the world and compare it with other composite indices. ADII (ASEAN Digital Integration Index) was proposed in 2021 as a way to assess digital progress in the South-East Asia after the COVID-19 pandemic. It consists of six pillars, five indicators each, focusing on digitally-enabled trade, cybersecurity, digital payment and identities, digital skills, innovation and entrepreneurship, and the readiness of institutions and infrastructure. The ADII was originally computed for 16 countries (ASEAN Member States and a few other economies as a benchmark). However, it is based on open sources with a global reach, allowing for the reconstruction of the index on a wider scale.

Using ADII 2.0 from 2023 as a guide, I reconstruct the scores for 108 countries from all over the world. For the countries included in the ADII 2.0 report, the index aligned with the original scores with 0.98 correlation and 7.58% average error. However, particular pillars were reconstructed with a lower precision, and the reconstructed scores tended to underestimate the ones from the report. The error rate was particularly large in case of smaller economies, such as Laos and Cambodia.

The comparison with other composite indices showed promising results. The highest correlation was obtained between ADII and the Tufts University's Digital Intelligence Index (Pearson: 0.94, Spearman: 0.97), quite good with the EU's Digital Economy and Society Index (0.87 and 0.89), and the lowest with ITU's Digitalisation Index (0.83 and 0.87). Spearman's correlations were always higher, meaning that the studies based on ranking countries are less affected by the choice of a particular digitalisation index.

Finally, I followed the approach proposed by Bruno et al. (2023) to reduce the dimensionality of the reconstructed index. Dimensionality reduction allows to lower the number of indicators, especially the ones with more missing data, and thus increase the geographical coverage. I managed to reduce the number of indicators from 30 to 24 and add 16 additional countries to the dataset, with 0.99 correlation with the full version, and 0.95 with the original ADII from the ASEAN's report. However, when analysed at the level of pillars, the quality of approximation was less encouraging.

The full dataset with reconstructed scores, as well as the version with reduced dimensionality, are available in the open repository. They can prove useful for the researchers comparing digitalisation level for a wide array of countries or performing spatial analysis of the economic development. Although the discontinuation of some of the data sources hinder the expansion of the dataset with more up-to-date indicators, the high similarity between composites of different indicators and the promising results of dimensionality reduction may allow future studies to replace missing variables with close alternatives, or remove them from the index altogether, without a substantial loss of quality.

Disclosures

This contribution is the result of the research project co-financed by the European Union under the Horizon Europe Framework Programme (HORIZON-MSCA-2021-SE-01-1; Project number: 101086381; Project title: Overcoming Digital Divide in Europe and Southeast Asia") and Polish Ministry of Science and Higher Education under Program "Projects International Co-financed“ ("Projekty Międzynarodowe Współfinansowane"; agreement number: 5319/HE/2022/2023/2)

Data availability

Full dataset is available in the Zenodo repository at `https://doi.org/10.5281/zenodo.21216158`

The code in R used to reconstruct the scores and perform the analysis can be found in the GitHub repository at `https://github.com/agachocz/ADII.git`

Literatura

ASEAN. ASEAN Digital Integration Index 2.0. Technical report, 2023.

Margarita Billon, Fernando Lera-Lopez, and Rocío Marco. Differences in digitalization levels: A multivariate analysis studying the global digital divide. *Review of World Economics*, 146(1):39–73, 3 2010. ISSN 16102878. doi: 10.1007/s10290-009-0045-y.

Ryszard Borowiecki, Barbara Siuta-tokarska, Jolanta Maroń, Marcin Suder, Agnieszka Thier, and Katarzyna Żmija. Developing digital economy and society in the light of the issue of digital convergence of the markets in the european union countries. *Energies*, 14(9), 5 2021. ISSN 19961073. doi: 10.3390/en14092717.

Giuseppe Bruno, Antonio Diglio, Carmela Piccolo, and Eduardo Pipicelli. A reduced Composite Indicator for Digital Divide measurement at the regional level: An application to the Digital Economy and Society Index (DESI). *Technological Forecasting and Social Change*, 190, 5 2023. ISSN 00401625. doi: 10.1016/j.techfore.2023.122461.

Noelia Cámara and David Tuesta. DiGiX: The Digitization Index. Technical report, 2017. URL `www.bbvaresearch.com`.

Bhaskar Chakravorti, Ravi Shankar Chaturvedi, Christina Filipovic, and Griffin Brewer. Digital in the Time of Covid. Trust in the Digital Economy and Its Evolution Across 90 Economies as the Planet Paused for a Pandemic. Technical report, Fletcher School at Tufts University, 12 2020.

Agnieszka Choczyńska, Septia Rani, and Justyna Tora. Comparison of international digitalisation indexes: A quantitative analysis perspective. *International Entrepreneurship Review*, 10(2):25–42, 6 2024. doi: 10.15678/ier.2024.1002.02.

European Commission. Digital Economy and Society Index (DESI) 2022 Thematic chapters. Technical report, European Commission, 2022a.

European Commission. Digital Economy and Society Index (DESI) 2022 Methodological Note. Technical report, 2022b.

Barbara Fura, Aneta Karasek, and Beata Hysa. Statistical assessment of digital transformation in European Union countries under sustainable development goal 9. *Quality and Quantity*, 59(1):937–972, 2 2025. ISSN 15737845. doi: 10.1007/s11135-024-01972-0.

International Telecommunication Union. Measuring digital development. The ICT Development Index 2023. Technical report, ITU, 2023.

Milica Jovanović, Jasmina Dlačić, and Milan Okanović. Digitalization and society's sustainable development – Measures and implications. *Zbornik Radova Ekonomskog Fakultet au Rijeci*, 36(2):905–928, 12 2018. ISSN 18467520. doi: 10.18045/zbefri.2018.2.905.

Olena Korzhyk, Jorge Vareda Gomes, and Gonçalo João. A Comparative Study of Different Digitalization Indexes. pages 238–267. 9 2023. doi: 10.4018/978-1-6684-9151-5.ch015.

Marcin Kotarba. Measuring Digitalization-Key Metrics. *Foundations of Management*, 9(1): 123–138, 2 2017. ISSN 23005661. doi: 10.1515/fman-2017-0010.

Tünde Zita Kovács, András Nábrádi, and Beáta Bittner. Digital Technology Integration Among Eastern European Companies, Based on Digital Economy and Society Index. *Interdisciplinary Description of Complex Systems*, 21(5):421–440, 2023. ISSN 1334-4684. doi: 10.7906/indecs.21.5.1.

Fei Li. Disconnected in a pandemic: COVID-19 outcomes and the digital divide in the United States. *Health and Place*, 77, 9 2022. ISSN 18732054. doi: 10.1016/j.healthplace.2022.102867.

Shu Chun Liao, Tzu Chuan Chou, and Chen Hao Huang. Revisiting the development trajectory of the digital divide: A main path analysis approach. *Technological Forecasting and Social Change*, 179, 6 2022. ISSN 00401625. doi: 10.1016/j.techfore.2022.121607.

Sophie Lythreatis, Sanjay Kumar Singh, and Abdul Nasser El-Kassar. The digital divide: A review and future research agenda. *Technological Forecasting and Social Change*, 175, 2 2022. ISSN 00401625. doi: 10.1016/j.techfore.2021.121359.

Farooq Mubarak, Reima Suomi, and Satu Päivi Kantola. Confirming the links between socio-economic variables and digitalization worldwide: the unsettled debate on digital divide. *Journal of Information, Communication and Ethics in Society*, 18(3):415–430, 8 2020. ISSN 17588871. doi: 10.1108/JICES-02-2019-0021.

Godwin Myovella, Mehmet Karacuka, and Justus Haucap. Digitalization and economic growth: A comparative analysis of Sub-Saharan Africa and OECD economies. *Telecommunications Policy*, 44(2), 3 2020. ISSN 03085961. doi: 10.1016/j.telpol.2019.101856.

Abdulkarim A. Oloyede, Nasir Faruk, Nasir Noma, Ebinimi Tebepah, and Augustine K. Nwaulune. Measuring the impact of the digital economy in developing countries: A systematic review and meta- analysis. *Heliyon*, 9(7), 7 2023. ISSN 24058440. doi: 10.1016/j.heliyon.2023.e17654.

Christopher Perumal, Novel Lyndon, and Zaini Sakawi. Evaluating ICT development: Indicators, digital progress, and societal impact for sustainable development. *International Journal of Advanced and Applied Sciences*, 11(9):36–47, 9 2024. ISSN 23133724. doi: 10.21833/ijaas.2024.09.005.

Vience Mutiara Rumata and Ashwin Sasongko Sastrosubroto. The Paradox of Indonesian Digital Economy Development. In *E-Business - Higher Education and Intelligence Applications*. IntechOpen, 2020. URL `www.intechopen.com`.

Araba Sey. Gender Digital Equality Across ASEAN. *ERIA Discussion Paper Series*, 358, 2021.

Barbara Siuta-Tokarska, Iwan Awaluddin Yusuf, Malgorzata Kowalik, Andrej Privara, and Marcin Suder. Working paper The Assessment of the State of the Digital Economy in South-East Asia; Project title: Overcoming Digital Divide in Europe and Southeast Asia. Working paper WP2 2. Technical report.

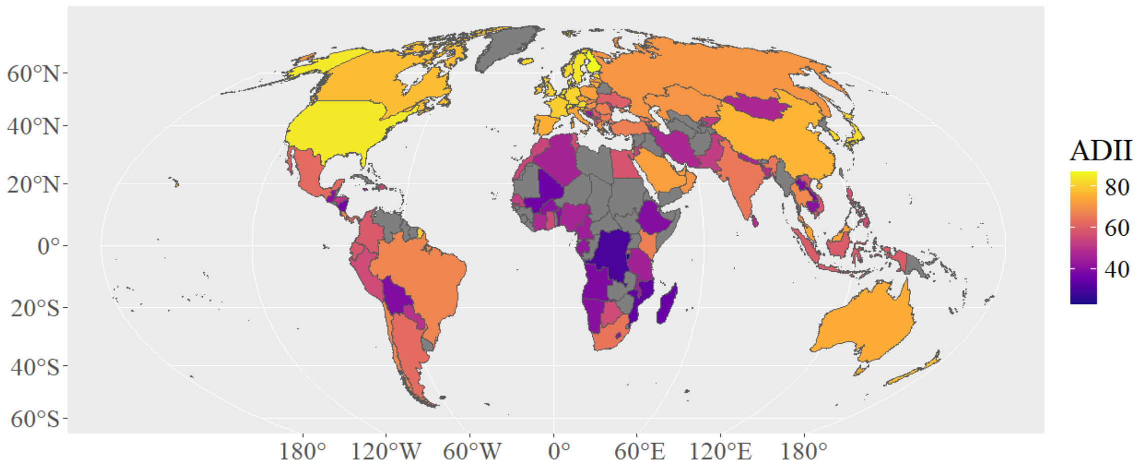
Gregory Vial. Understanding digital transformation: A review and a research agenda, 6 2019. ISSN 09638687.

Katarzyna Żmija, Wahida Ahmad, Wojciech Szymla, and Marcin Suder. The Assessment of the State of Digital Economy in Selected EU Countries; Project title: Overcoming Digital Divide in Europe and Southeast Asia. Working paper WP2 2. Technical report.

Appendix

Pillar	Correlation	Average pct. error	Average error
P1	0.89	16.12	-7.38
P2	0.90	11.71	1.20
P3	0.78	15.81	2.61
P4	0.78	24.06	11.53
P5	0.83	17.88	4.99
P6	0.94	7.48	1.21
Index	0.95	7.06	2.36

Tabela 4: Comparison of reduced pillar scores with original values from the ADII report (ASEAN, 2023) N = 14.



Rysunek 2: Reduced version of the ADII scores. Countries with unavailable data are in grey,